

My Name : Haya Nahhas

Assignment Title : QSAR Data Curation Using ChEMBL

###Introduction###

PI3-kinase class I is a key target in breast cancer, as it is involved in activating cellular signaling pathways for cancer cell growth. In this assignment, bioactivity data from ChEMBL is curated and analyzed for responsible use in QSAR modeling, aiming to enhance the prediction of the effects of drugs inhibiting this target. This analysis will contribute to the development of more precise and effective drugs for breast cancer treatment.

Target Name : PI3-kinase class I

:Background on PI3-kinase class I and Why It Was Chosen

PI3-kinase class I is a key enzyme in the PI3K/AKT/mTOR pathway that regulates cell growth and survival. In breast cancer, excessive activation of this pathway leads to increased tumor growth and resistance to treatment. This target was chosen because mutations in the PIK3CA gene occur in a significant proportion of breast cancer cases, especially in triple-negative breast cancer, which increases enzyme activity and contributes to cancer spread.

There are 432 bioactivity records available from the ChEMBL database for PI3-kinase class I, making it suitable for QSAR modeling and the development of precise drugs to improve breast cancer treatment.

Data Retrieval and Processing Steps :

-Data Retrieval: Bioactivity data for PI3-kinase class I was retrieved from the ChEMBL database using Python and the ChEMBL webservice. IC50 was selected as the primary measure of compound efficacy.

-Data Cleaning: The data was examined, and records containing missing or invalid values were removed to ensure data accuracy and quality.

-Measurement Standardization: IC50 measurements were standardized to a consistent unit (e.g., micromolar) to ensure uniformity across records.

-Data Saving: The formatted data was saved into a CSV file for use in QSAR modeling.

Workflow for Data Curation Using ChEMBL

Automation & Data Retrieval:

Instead of manually downloading the data, we used the ChEMBL web resource client in Python to save time and work directly with the code.

The notebook was imported directly from the official AI & Biotechnology GitHub repository, and we created a copy to work on.

Collecting Data in Google Colab:

We linked Google Colab with Google Drive and created a folder named "data" to store all CSV files.

We began by installing the necessary libraries such as ChEMBL web resource client and Pandas.

Code :

```
!pip install chembl_webresource_client
```

After installation, the libraries were imported: ChEMBL client for data retrieval and Pandas to organize tables and store data as DataFrames.

Code:

```
import pandas as pd
from chembl_webresource_client.new_client import new_client
```

Target Selection:

The selected target is PI3-kinase class I due to its significant role in breast cancer and its relevance for QSAR modeling. The search results were converted into a DataFrame using Pandas, as the data returned by the platform was not formatted into a table.

Code :

```
target = new_client.target
target_query = target.search("PI3-kinase class I")
targets = pd.DataFrame.from_dict(target_query)
targets.head()
```

Bioactivity Data Processing:

We focused solely on IC50 values to avoid inconsistencies between values such as ki and kd, ensuring the reliability of the modeling. We applied filters to retrieve the correct data:

Filter 1: Filter by the target identifier, **Filter 2:** Filter bioactivity records to retain only IC50 values.

Code:

```
activity = new_client.activity
results =
activity.filter(target_chembl_id=selected_target).filter(standard_type="IC50")
```

Saving Data:

The raw bioactivity data was saved as a CSV file for further processing later.

Data Quality Assurance:

Before model building, we ensured that all compounds in the dataset had valid IC50 values

Code:

```
df1["standard_type"].isna().sum()
```

Any compound missing an IC50 value was excluded from the dataset to avoid confusion in the model

Code:

```
df2 = df1[df1["standard_value"].notna()]
df2.head()
```

Bioactivity Classification:

We classified the compounds into three categories based on their IC50 values:

Active: Compounds with $IC_{50} \leq 1000$ nM.

Inactive: Compounds with $IC_{50} > 10000$ nM.

Moderate Activity: Compounds with IC50 between 1000 and 10000 nM.

We created an empty list named bioactivity class to store the bioactivity values for each compound.

A for loop was used to classify each IC50 value into the corresponding bioactivity category.

We extracted only the necessary columns, such as Molecule-ids, Canonical-smiles, and Standard-values

Code :

```
bioactivity_class = []
for value in df2.standard_value:
    value = float(value)
    if value >= 10000:
        bioactivity_class.append("inactive")
    elif value <= 1000:
        bioactivity_class.append("active")
    else:
        bioactivity_class.append("intermediate")
molecule_ids = df2.molecule_chembl_id.tolist()
canonical_smiles = df2.canonical_smiles.tolist()
standard_values = df2.standard_value.tolist()
```

Merging Data:

The bioactivity classifications were merged with the relevant compound columns.

Removing Missing Values:

We ensured that records with missing smiles values were removed, as compounds without these values cannot have their properties calculated

Code :

```
df3 = df3.dropna(subset=["canonical_smiles"])
df3 = df3[df3["canonical_smiles"].str.lower() != "none"]
df3 = df3[df3["canonical_smiles"].str.strip() != ""]
df3.head()
```

🚀 **Saving Final Data:**

The final curated and classified data was saved to Google Drive in the "data" folder, The files resulting from this process were: Bioactivity data, Raw bioactivity data

Code :

```
df3.to_csv("bioactivity_preprocessed_data.csv", index=False)

!cp bioactivity_preprocessed_data.csv "/content/gdrive/My Drive/Colab Notebooks/data"
!ls "/content/gdrive/My Drive/Colab Notebooks/data"
```

My repository link:

https://github.com/Haya-biotech/AI_and_drug_discovery_course_2026.git